

R PROGRAMMING LAB

PRACTICAL NO 6

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Date of Practical: 08/09/2026	Date of Submission: 12/09/2026

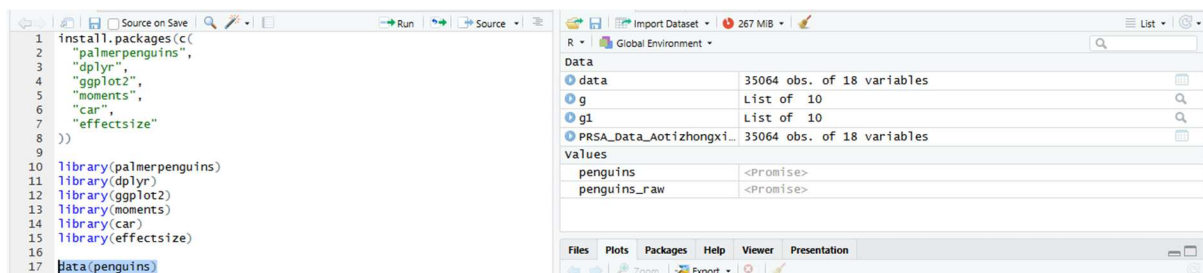
Lab Problem Statement 6: Statistical Analysis of Physical Characteristics of Palmer Penguins

Problem Statement:

Researchers studying penguins in the Palmer Archipelago, Antarctica, collected measurements from different penguin species to understand variations in their physical characteristics.

Using the Palmer Penguins dataset, perform a comprehensive statistical analysis in R to investigate whether physical characteristics such as body mass and flipper length differ significantly across penguin species and sex. The analysis should combine descriptive statistics, hypothesis testing, effect-size estimation, ANOVA, post-hoc analysis, and appropriate non-parametric tests.

Dataset: UCI Machine Learning Repository – Palmer Penguins Dataset



```
17 data(penguins)
18 head(penguins)
```

18:1 (Top Level) R Script

Console Terminal Background Jobs

R 4.6.1 ~/

```
> library(effects)
> data(penguins)
> head(penguins)
# A tibble: 6 × 8
  species island bill_length_mm bill_depth_mm flipper_length_mm
  <fct>   <fct>         <dbl>         <dbl>         <int>
1 Adelie  Torgersen      39.1           18.7           181
2 Adelie  Torgersen      39.5           17.4           186
3 Adelie  Torgersen      40.3            18           195
4 Adelie  Torgersen      NA             NA             NA
5 Adelie  Torgersen      36.7           19.3           193
6 Adelie  Torgersen      39.3           20.6           190
# i 3 more variables: body_mass_g <int>, sex <fct>, year <int>
> |
```

```
> dim(penguins)
[1] 344  8
```

```
> summary(penguins)
```

species	island	bill_length_mm	bill_depth_mm
Adelie :152	Biscoe :168	Min. :32.10	Min. :13.10
Chinstrap: 68	Dream :124	1st Qu.:39.23	1st Qu.:15.60
Gentoo :124	Torgersen: 52	Median :44.45	Median :17.30
		Mean :43.92	Mean :17.15
		3rd Qu.:48.50	3rd Qu.:18.70
		Max. :59.60	Max. :21.50
		NAS :2	NAS :2

flipper_length_mm	body_mass_g	sex	year
Min. :172.0	Min. :2700	female:165	Min. :2007
1st Qu.:190.0	1st Qu.:3550	male :168	1st Qu.:2007
Median :197.0	Median :4050	NAS : 11	Median :2008
Mean :200.9	Mean :4202		Mean :2008
3rd Qu.:213.0	3rd Qu.:4750		3rd Qu.:2009
Max. :231.0	Max. :6300		Max. :2009
NAS :2	NAS :2		

```
> |
```

PART A — DESCRIPTIVE STATISTICAL ANALYSIS

Descriptive statistics are statistical measures used to summarize and describe the important characteristics of a dataset. They provide information about the central tendency, dispersion, distribution and shape of the data.

1. Overall Descriptive Statistics

```
> dim(penguins_clean)
[1] 333    8
> mean(penguins_clean$body_mass_g)
[1] 4207.057
> median(penguins_clean$body_mass_g)
[1] 4050
> min(penguins_clean$body_mass_g)
[1] 2700
> max(penguins_clean$body_mass_g)
[1] 6300
> var(penguins_clean$body_mass_g)
[1] 648372.5
> sd(penguins_clean$body_mass_g)
[1] 805.2158
> |
```

2. Quartiles and IQR

```
> quantile(
+   penguins_clean$body_mass_g,
+   probs = c(0.25, 0.50, 0.75)
+ )
 25%  50%  75%
3550 4050 4775
> IQR(penguins_clean$body_mass_g)
[1] 1225
> |
```

3. Skewness

```
> skewness(penguins_clean$body_mass_g)
[1] 0.4701162
|
```

Interpretation

- Skewness $\approx 0 \rightarrow$ approximately symmetric
- Positive $\rightarrow$ right-skewed
- Negative $\rightarrow$ left-skewed

4. Kurtosis

```
> kurtosis(penguins_clean$body_mass_g)
[1] 2.259514
|
```

Interpretation:

- Higher kurtosis $\rightarrow$ heavier tails/more extreme observations
- Lower kurtosis $\rightarrow$ lighter tails
- Around 3 $\rightarrow$ approximately normal kurtosis when using the moments::kurtosis() definition

```
> descriptive_stats
      Mean Median Minimum Maximum Variance      SD   Q1   Q3  IQR
25% 4207.057  4050    2700    6300 648372.5 805.2158 3550 4775 1225
      Skewness Kurtosis
25% 0.4701162 2.259514
> |
```

5. Species-Wise Descriptive Statistics

Species-wise descriptive analysis calculates statistical measures separately for Adélie, Chinstrap and Gentoo penguins. This allows us to compare their physical characteristics and identify differences in their central tendency and variability.

```
> species_stats
# A tibble: 3 × 12
  species Mean Median Minimum Maximum Variance SD Q1 Q3
  <fct>   <dbl> <dbl> <int> <int> <dbl> <dbl> <dbl> <dbl>
1 Adélie 3706. 3700 2850 4775 210332. 459. 3362. 4000
2 Chinstrap 3733. 3700 2700 4800 147713. 384. 3488. 3950
3 Gentoo 5092. 5050 3950 6300 251478. 501. 4700 5500
# i 3 more variables: IQR <dbl>, skewness <dbl>, kurtosis <dbl>
> |
```

The Gentoo penguins have a substantially higher average body mass than both Adélie and Chinstrap penguins.

Adélie and Chinstrap have relatively similar mean body masses, whereas Gentoo shows a clear increase. This difference is also visible in the species-wise boxplot.

Interpretation

The average body mass of the penguins is approximately 4207 g, while the median body mass is 4050 g. The mean being slightly higher than the median indicates some higher body-mass observations influencing the average.

The body mass ranges from 2700 g to 6300 g, giving a wide range of 3600 g. The standard deviation of approximately 805 g indicates considerable variation in body mass among the penguins.

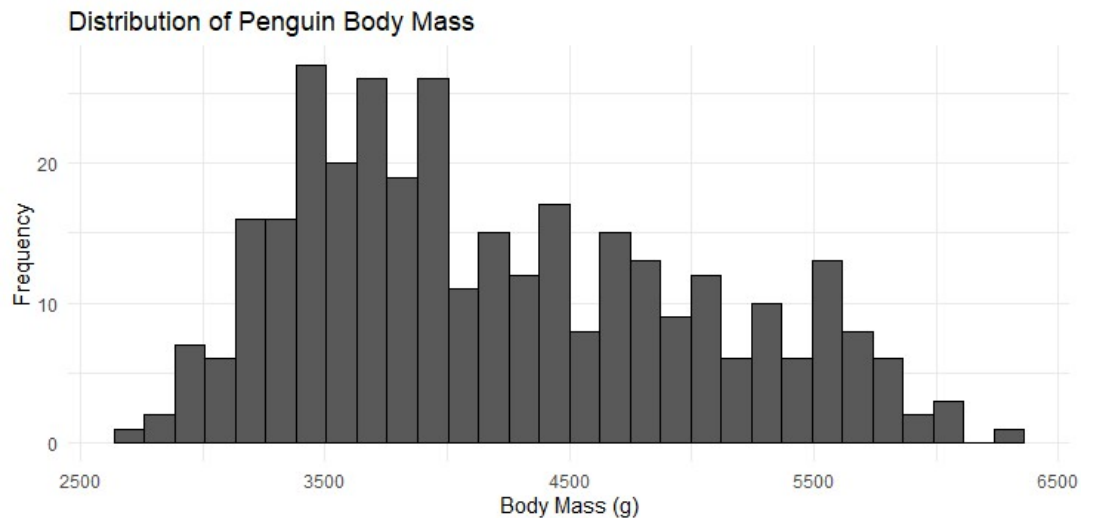
The first quartile is 3550 g and the third quartile is 4775 g, giving an IQR of 1225 g. Therefore, the middle 50% of penguins have body masses between approximately 3550 g and 4775 g.

The skewness is 0.4701, which is positive. Therefore, the distribution is moderately right-skewed, meaning there is a somewhat longer tail toward higher body-mass values.

The kurtosis obtained is 2.2595. Since it is below 3 using the `moments::kurtosis()` definition, the distribution has relatively lighter tails than a normal distribution.

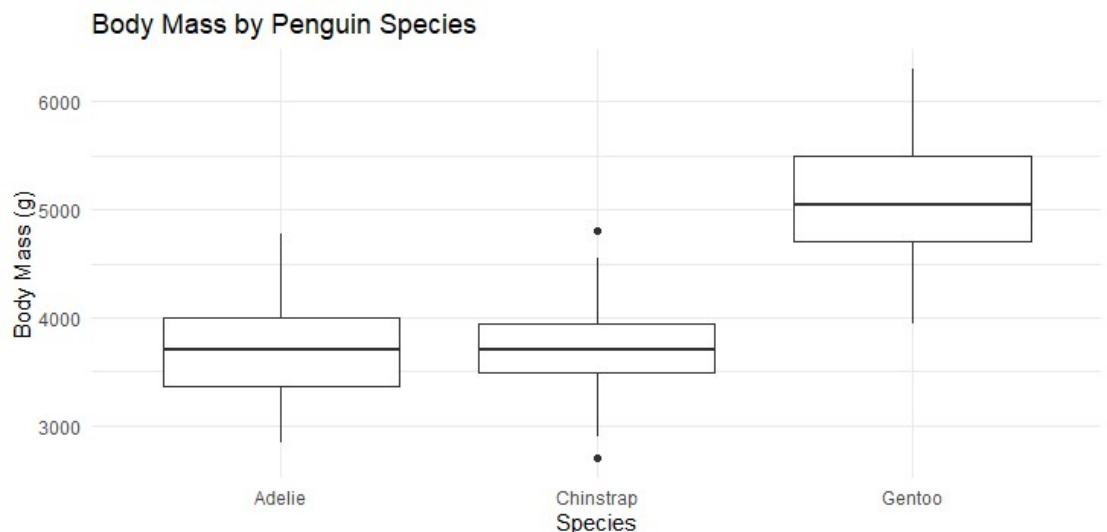
PART B — VISUALIZATION

1. Histogram of Body Mass



The histogram of body mass shows that the observations are spread over approximately **2700–6300 g**. The distribution is not perfectly symmetric, which agrees with the positive skewness value of **0.4701**.

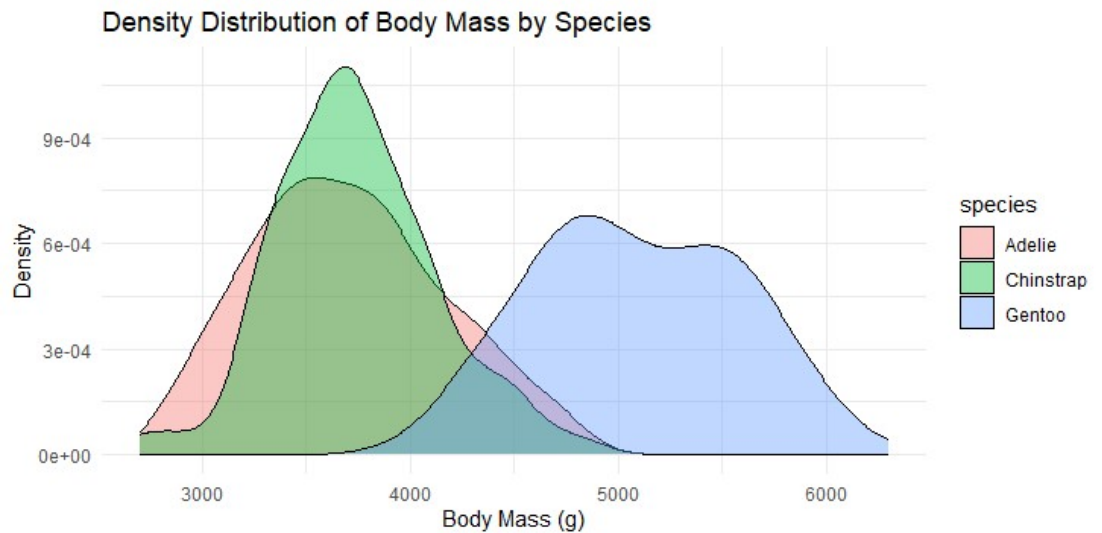
2. Species-Wise Boxplot



The boxplot shows a clear difference in body mass among species. Gentoo penguins have considerably higher body masses, while Adélie and Chinstrap penguins have similar distributions.

3. Density Plot

The density plot shows that the distributions of body mass differ across species. The Gentoo distribution is shifted toward higher body-mass values, while Adélie and Chinstrap distributions occur at lower and relatively similar ranges.



PART C — HYPOTHESIS TESTING

A two-sample independent t-test determines whether the means of two independent groups are significantly different.

Here, the two groups are:

- Female penguins
- Male penguins

Null Hypothesis — H_0

There is no significant difference between the mean body masses of male and female penguins.

$$H_0: \mu_{male} = \mu_{female}$$

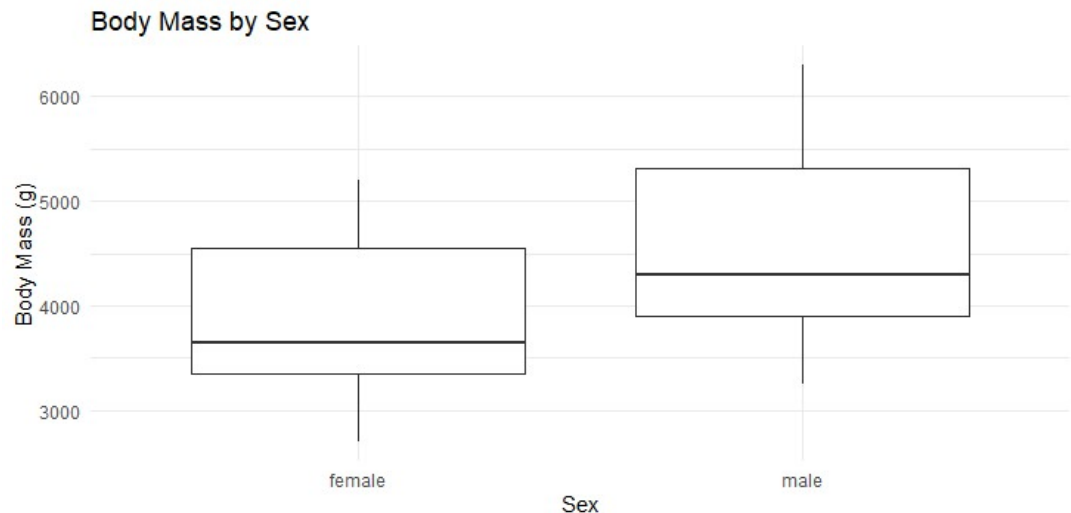
Alternative Hypothesis — H_1

There is a significant difference between the mean body masses of male and female penguins.

$$H_1: \mu_{male} \neq \mu_{female}$$

This is a two-tailed test.

1. Visualize Male vs Female Body Mass



2. Check Normality — Shapiro-Wilk Test

shapiro-wilk normality test

```
data: penguins_clean$body_mass_g[penguins_clean$sex == "male"]
w = 0.92504, p-value = 1.227e-07
```

```
> shapiro.test(
+   penguins_clean$body_mass_g[
+     penguins_clean$sex == "female"
+   ]
+ )
```

shapiro-wilk normality test

```
data: penguins_clean$body_mass_g[penguins_clean$sex == "female"]
w = 0.91931, p-value = 6.155e-08
```

Male

- $W = 0.92504$
- $p\text{-value} = 1.227 \times 10^{-7}$

Female

- $W = 0.91931$
- $p\text{-value} = 6.155 \times 10^{-8}$

Both p-values are much smaller than 0.05.

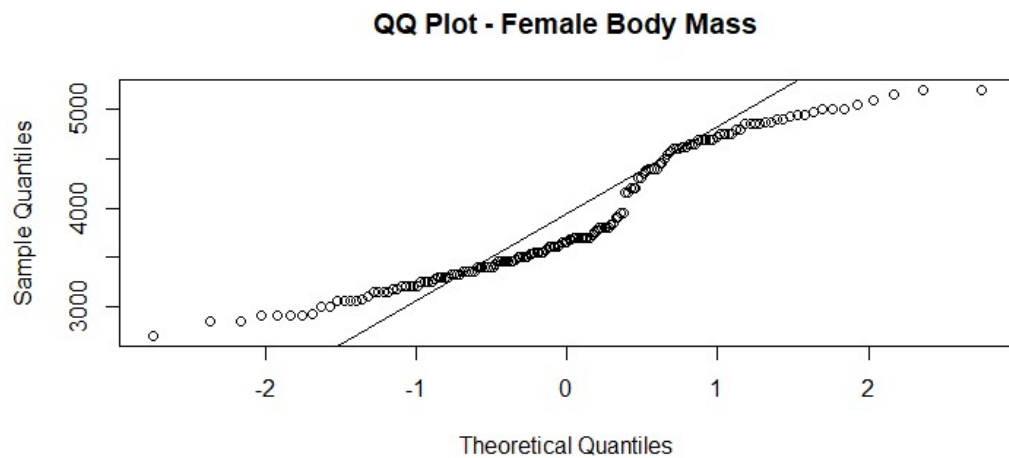
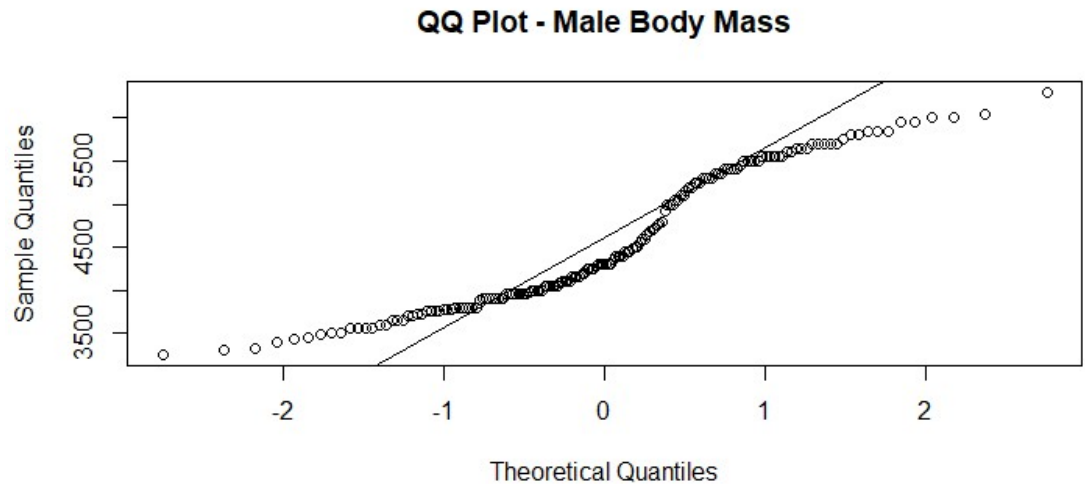
Interpretation

Since both p-values are less than 0.05, the null hypothesis of normality is rejected for both male and female body-mass samples.

Therefore, according to the Shapiro-Wilk test, the distributions significantly deviate from normality.

However, the QQ plots should also be considered when assessing the practical severity of this deviation. The practical itself specifies using both Shapiro-Wilk testing and QQ plots for normality assessment.

3. QQ Plot for Normality



Interpretation:

If points approximately follow the straight line, the data is reasonably close to normal.

4. Independent Two-Sample t-Test

The independent two-sample t-test determines whether the means of two independent groups differ significantly.

```
> t_test_result

welch Two Sample t-test

data:  body_mass_g by sex
t = -8.5545, df = 323.9, p-value = 4.794e-16
alternative hypothesis: true difference in means between group female and
group male is not equal to 0
95 percent confidence interval:
 -840.5783 -526.2453
sample estimates:
mean in group female    mean in group male
      3862.273           4545.685
```

Result

Your output gives:

- $t = -8.5545$
- $df = 323.9$
- $p\text{-value} = 4.794 \times 10^{-16}$

Mean body masses:

- Female = 3862.273 g
- Male = 4545.685 g

Interpretation

The p-value is far below 0.05.

Therefore, we reject H_0 and conclude that there is a statistically significant difference between the mean body masses of male and female penguins.

The mean body mass of males is approximately 683 g higher than that of females.

Thus, sex is strongly associated with body mass in this dataset.

5. Extract the Confidence Interval

```
> t_test_result$conf.int
[1] -840.5783 -526.2453
attr(,"conf.level")
[1] 0.95
> |
```

Interpretation

We are 95% confident that the true difference between the female and male mean body masses lies between approximately -840.58 g and -526.25 g.

Because the entire interval is below zero, it supports the conclusion that females have a lower average body mass than males.

Importantly, the confidence interval does not contain zero, which is consistent with the statistically significant t-test.

6. Calculate Cohen's d

Cohen's d measures the standardized difference between two group means. Unlike the p-value, which tells us whether a difference is statistically significant, Cohen's d tells us about the magnitude of the difference

```

> cohens_d <- cohens_d(
+   body_mass_g ~ sex,
+   data = penguins_clean
+ )
> cohens_d
Cohen's d |          95% CI
-----|-----
-0.94    | [-1.16, -0.71]

- Estimated using pooled SD.
> |

```

Interpretation

The absolute value of Cohen's d is approximately 0.94, which represents a large effect size.

Therefore, the difference in body mass between male and female penguins is not only statistically significant but also practically substantial.

PART D — ONE-WAY ANOVA

One-way ANOVA (Analysis of Variance) is used to compare the means of three or more independent groups.

Here, we compare body mass among:

- Adélie
- Chinstrap
- Gentoo

H_0

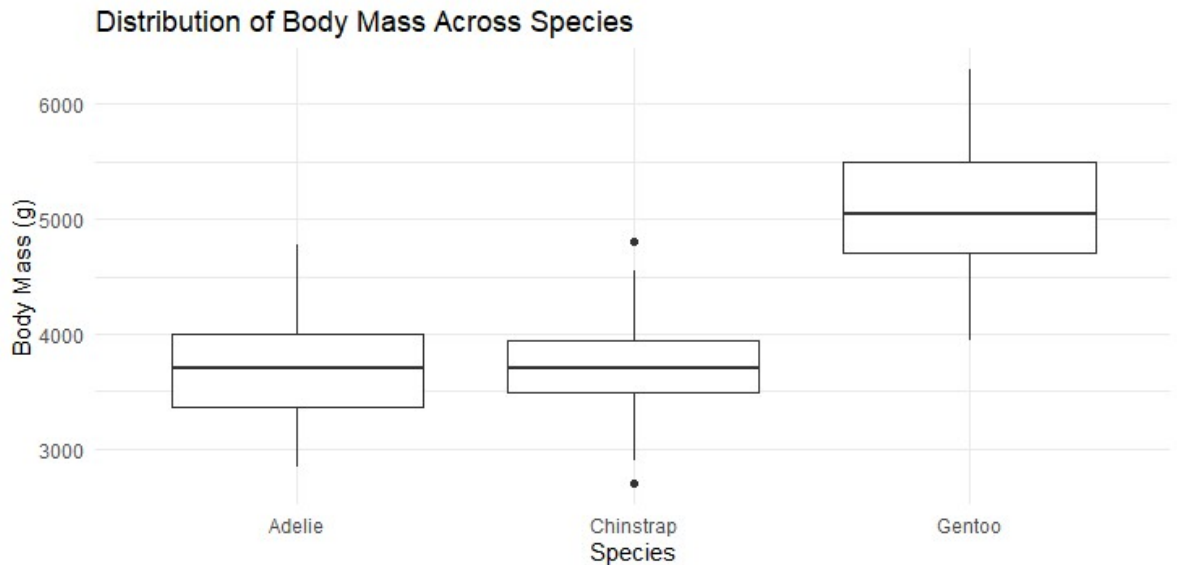
The mean body mass is equal across all three species.

$$\mu_{Adelie} = \mu_{Chinstrap} = \mu_{Gentoo}$$

H_1

At least one species has a different mean body mass.

1. Examine Species Distributions



2. Check Normality Within Each Species

```
> shapiro.test(  
+   penguins_clean$body_mass_g[  
+     penguins_clean$species == "Adelie"  
+   ]  
+ )
```

shapiro-wilk normality test

```
data:  penguins_clean$body_mass_g[penguins_clean$species == "Adelie"]  
W = 0.98116, p-value = 0.04232
```

```
> shapiro.test(  
+   penguins_clean$body_mass_g[  
+     penguins_clean$species == "Chinstrap"  
+   ]  
+ )
```

shapiro-wilk normality test

```
data:  penguins_clean$body_mass_g[penguins_clean$species == "Chinstrap"]  
W = 0.98449, p-value = 0.5605
```

```
> shapiro.test(  
+   penguins_clean$body_mass_g[  
+     penguins_clean$species == "Gentoo"  
+   ]  
+ )
```

shapiro-wilk normality test

```
data:  penguins_clean$body_mass_g[penguins_clean$species == "Gentoo"]  
W = 0.98606, p-value = 0.2605
```

Interpretation

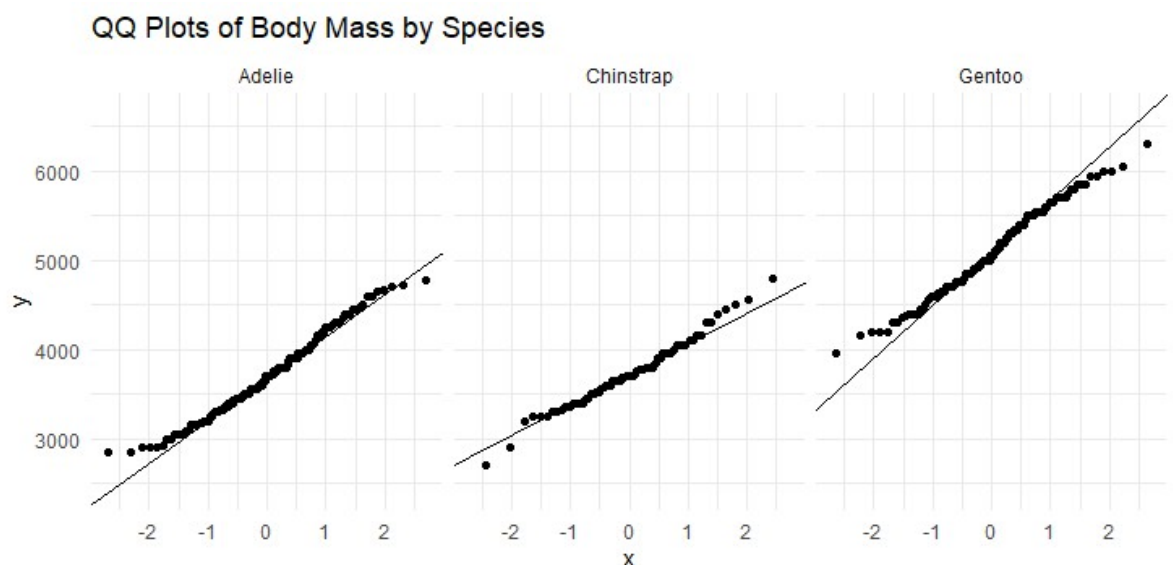
For Adélie, the p-value is 0.04232, which is slightly below 0.05. Therefore, the normality assumption is rejected for Adélie.

For Chinstrap and Gentoo, the p-values are greater than 0.05, so there is insufficient evidence to reject normality.

Thus, the normality assumption is slightly violated for Adélie, while Chinstrap and Gentoo do not show significant evidence of non-normality.

The QQ plots should be considered alongside these tests.

3. QQ Plots for All Species



4. Check Homogeneity of Variance

```
> leveneTest(
+   body_mass_g ~ species,
+   data = penguins_clean
+ )
Levene's Test for Homogeneity of Variance (center = median)
      Df F value    Pr(>F)
group  2  5.1349 0.006367 **
    330

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
> |
```

Result

- $F = 5.1349$
- $p\text{-value} = 0.006367$

Interpretation

Since:

$$p = 0.006367 < 0.05$$

we reject H_0 .

Therefore, the variances of body mass across the three species are significantly different.

This means the homogeneity-of-variance assumption for conventional one-way ANOVA is violated.

5. Perform One-Way ANOVA

```
> anova_model <- aov(
+   body_mass_g ~ species,
+   data = penguins_clean
+ )
>
> summary(anova_model)
```

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
species	2	145190219	72595110	341.9	<2e-16 ***
Residuals	330	70069447	212332		

```
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
> |
```

ANOVA output:

- $df = 2$
- $F = 341.9$
- $p\text{-value} < 2 \times 10^{-16}$

Interpretation

The p-value is much smaller than 0.05.

Therefore, we reject the null hypothesis.

There is a statistically significant difference in mean body mass among the three penguin species.

The very large F-statistic (341.9) indicates that variation between species means is much greater than the residual variation within species.

However, ANOVA alone does not tell us which species differ from each other.

For that, Tukey's HSD is required.

6. Tukey HSD Post-Hoc Test

Tukey's Honestly Significant Difference (HSD) test is a post-hoc test used after a significant ANOVA to determine which specific pairs of group means are significantly different.

```
> tukey_result
```

```
Tukey multiple comparisons of means  
95% family-wise confidence level
```

```
Fit: aov(formula = body_mass_g ~ species, data = penguins_clean)
```

```
$species
```

	diff	lwr	upr	p adj
Chinstrap-Adelie	26.92385	-132.3528	186.2005	0.916431
Gentoo-Adelie	1386.27259	1252.2897	1520.2554	0.000000
Gentoo-Chinstrap	1359.34874	1194.4304	1524.2671	0.000000

Interpretation

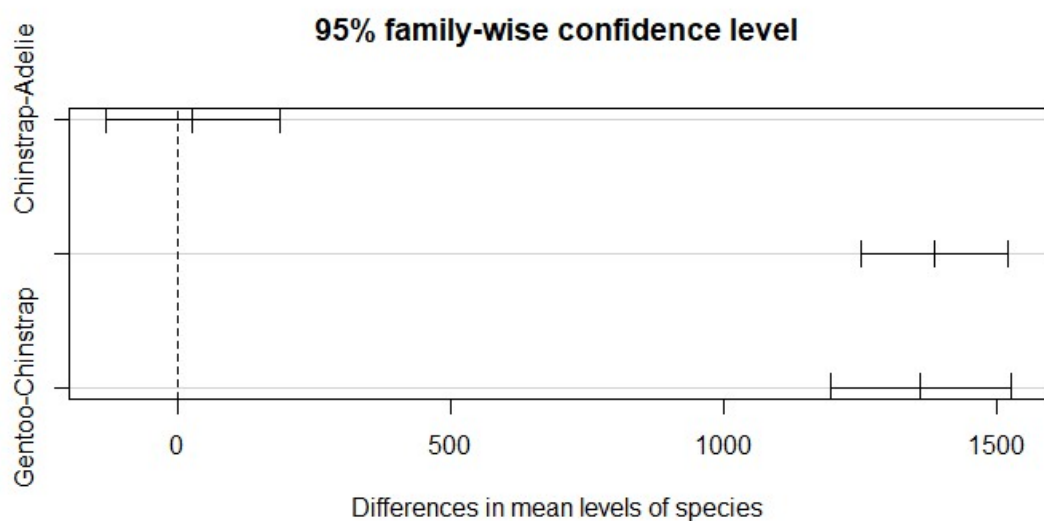
There is no significant difference between Adélie and Chinstrap mean body mass because:

$$p = 0.9164 > 0.05$$

However:

- Gentoo differs significantly from Adélie.
- Gentoo differs significantly from Chinstrap.

Therefore, the major species-level difference in body mass is due to Gentoo penguins having substantially greater body mass than the other two species.



PART E — KRUSKAL-WALLIS TEST

The Kruskal-Wallis test is a non-parametric alternative to one-way ANOVA. It is useful when the assumptions required for ANOVA, particularly normality or equal variance, are substantially violated.

```
> kruskal_result <- kruskal.test(  
+   body_mass_g ~ species,  
+   data = penguins_clean  
+ )  
>  
> kruskal_result  
  
      Kruskal-Wallis rank sum test  
  
data:  body_mass_g by species  
Kruskal-Wallis chi-squared = 212.09, df = 2, p-value <  
2.2e-16
```

Result

- $\chi^2 = 212.09$
- $df = 2$
- $p\text{-value} < 2.2 \times 10^{-16}$

Interpretation

The p-value is much smaller than 0.05.

Therefore, we reject H_0 and conclude that body mass differs significantly among penguin species.

PART F — TWO-WAY ANOVA

Two-way ANOVA investigates the effects of **two categorical independent variables** on a continuous dependent variable.

Here:

- Dependent variable → `body_mass_g`
- Factor 1 → `species`
- Factor 2 → `sex`

The model is:

$$\text{body_mass_g} \sim \text{species} * \text{sex}$$

The `*` also evaluates the interaction between species and sex.

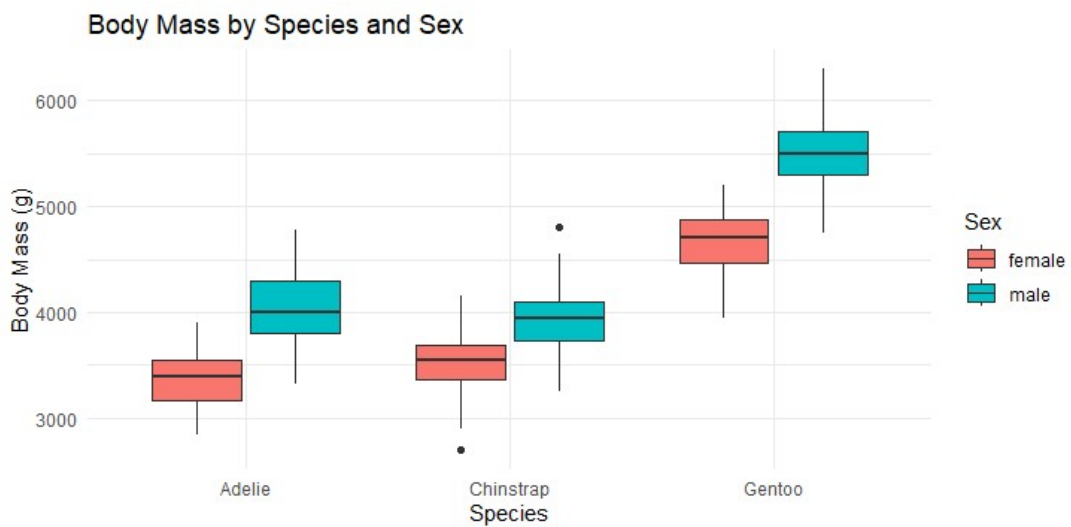
1. Create Two-Way ANOVA Model

```
> summary(two_way_model)
```

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
species	2	145190219	72595110	758.358	< 2e-16 ***
sex	1	37090262	37090262	387.460	< 2e-16 ***
species:sex	2	1676557	838278	8.757	0.000197 ***
Residuals	327	31302628	95727		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
> |

2. Visualize Species × Sex



Species Effect

Since:

$$p < 0.05$$

species has a statistically significant effect on body mass.

Sex Effect

Since:

$$p < 0.05$$

sex also has a statistically significant effect on body mass.

Interaction Effect

The interaction has:

$$p = 0.000197 < 0.05$$

Therefore, the interaction between species and sex is statistically significant.

Overall Interpretation

Both species and sex independently influence body mass, and the significant interaction indicates that the effect of sex is not exactly the same across all species.

In other words, differences between male and female body mass vary depending on the penguin species.

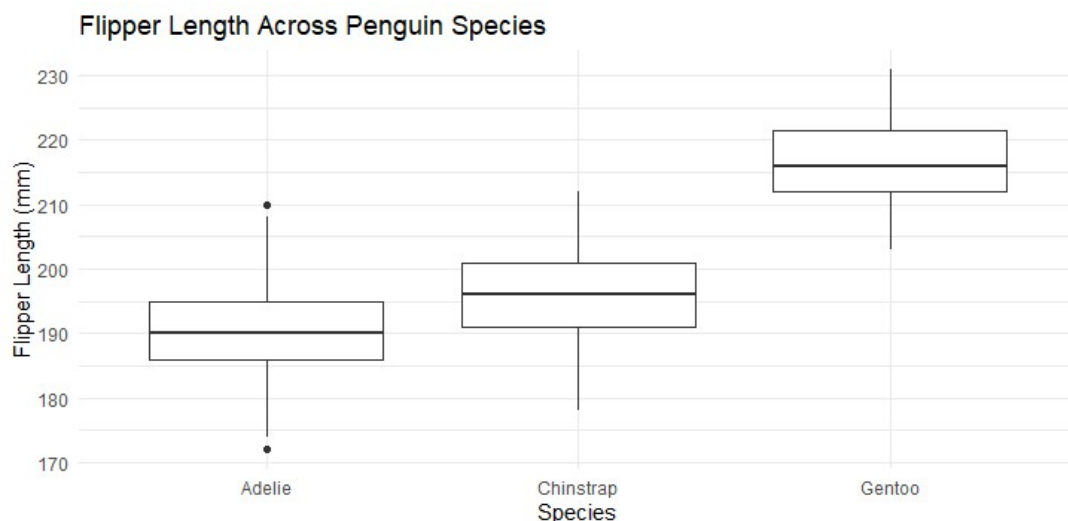
PART G — FLIPPER LENGTH ANALYSIS

Flipper length is another physical characteristic of penguins. A one-way ANOVA can be used to determine whether mean flipper length differs among Adélie, Chinstrap and Gentoo penguins.

1. Flipper Length Descriptive Statistics

```
> flipper_stats
# A tibble: 3 × 7
  species    Mean Median    SD Minimum Maximum   IQR
  <fct>    <dbl>  <dbl> <dbl>   <int>   <int> <dbl>
1 Adélie    190.    190   6.52    172    210     9
2 Chinstrap 196.    196   7.13    178    212    10
3 Gentoo    217.    216   6.59    203    231    9.5
> |
```

2. Flipper Length Boxplot



3. Flipper Length One-Way ANOVA

```
> flipper_anova <- aov(
+   flipper_length_mm ~ species,
+   data = penguins_clean
+ )
>
> summary(flipper_anova)
              Df Sum Sq Mean Sq F value Pr(>F)
species         2   50526    25263   567.4 <2e-16 ***
Residuals      330   14693         45
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
> |
```

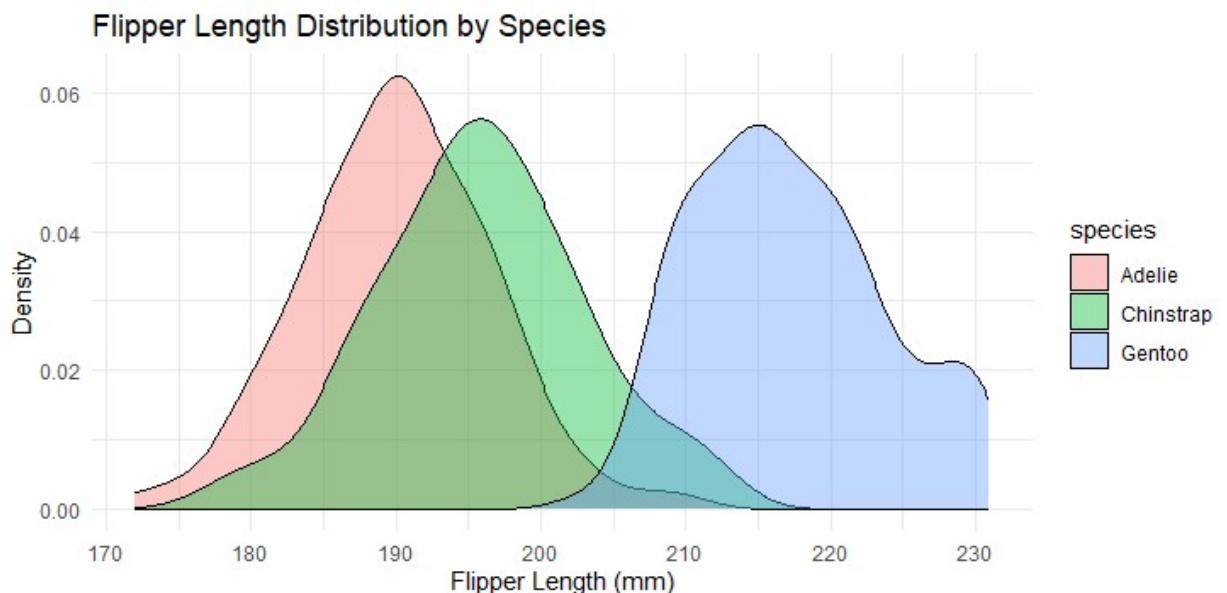
4. Tukey Test for Flipper Length

```
> TukeyHSD(flipper_anova)
Tukey multiple comparisons of means
95% family-wise confidence level
```

```
Fit: aov(formula = flipper_length_mm ~ species, data = penguins_clean)
```

```
$species
              diff      lwr      upr p adj
Chinstrap-Adelie  5.72079  3.414364  8.027215    0
Gentoo-Adelie     27.13255 25.192399 29.072709    0
Gentoo-Chinstrap 21.41176 19.023644 23.799885    0
```

5. Flipper Length Distribution



Interpretation

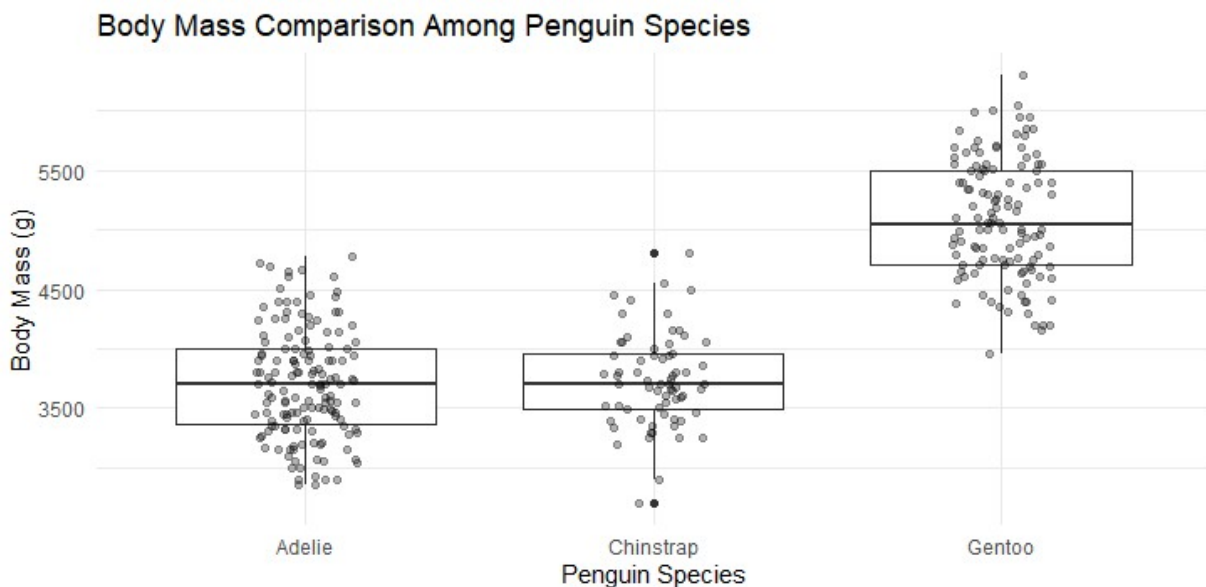
All three pairwise comparisons are statistically significant.

Therefore:

- Chinstrap flippers are significantly longer than Adélie flippers.
- Gentoo flippers are significantly longer than Adélie flippers.
- Gentoo flippers are significantly longer than Chinstrap flippers.

This is stronger than the body-mass result because all three species differ significantly in flipper length.

PART H — REQUIRED GROUP COMPARISON GRAPH



Overall Interpretation of the Experiment

The statistical analysis provides strong evidence that the physical characteristics of Palmer Penguins vary across species and sex.

The descriptive analysis showed substantial variation in body mass, with an overall mean of approximately 4207 g and positive skewness of 0.4701.

The independent t-test demonstrated a significant difference between male and female body masses. Males had an average body mass of approximately 4545.69 g, compared with 3862.27 g for females. The large Cohen's d of -0.94 indicates that this difference is also practically substantial.

The one-way ANOVA showed a highly significant difference in body mass among species. Tukey's test revealed that Gentoo differed significantly from both Adélie and Chinstrap, while the difference between Adélie and Chinstrap was not significant.

The assumptions for conventional ANOVA were not perfectly satisfied, as indicated by the significant Levene's test. Therefore, the Kruskal-Wallis test was also performed. It produced a highly significant result, supporting the same overall conclusion as ANOVA.

The two-way ANOVA further showed that species, sex and their interaction significantly affect body mass.

Finally, the flipper-length analysis showed that species also differ significantly in flipper length. Unlike body mass, all three species pairs showed significant differences in flipper length according to Tukey's HSD.